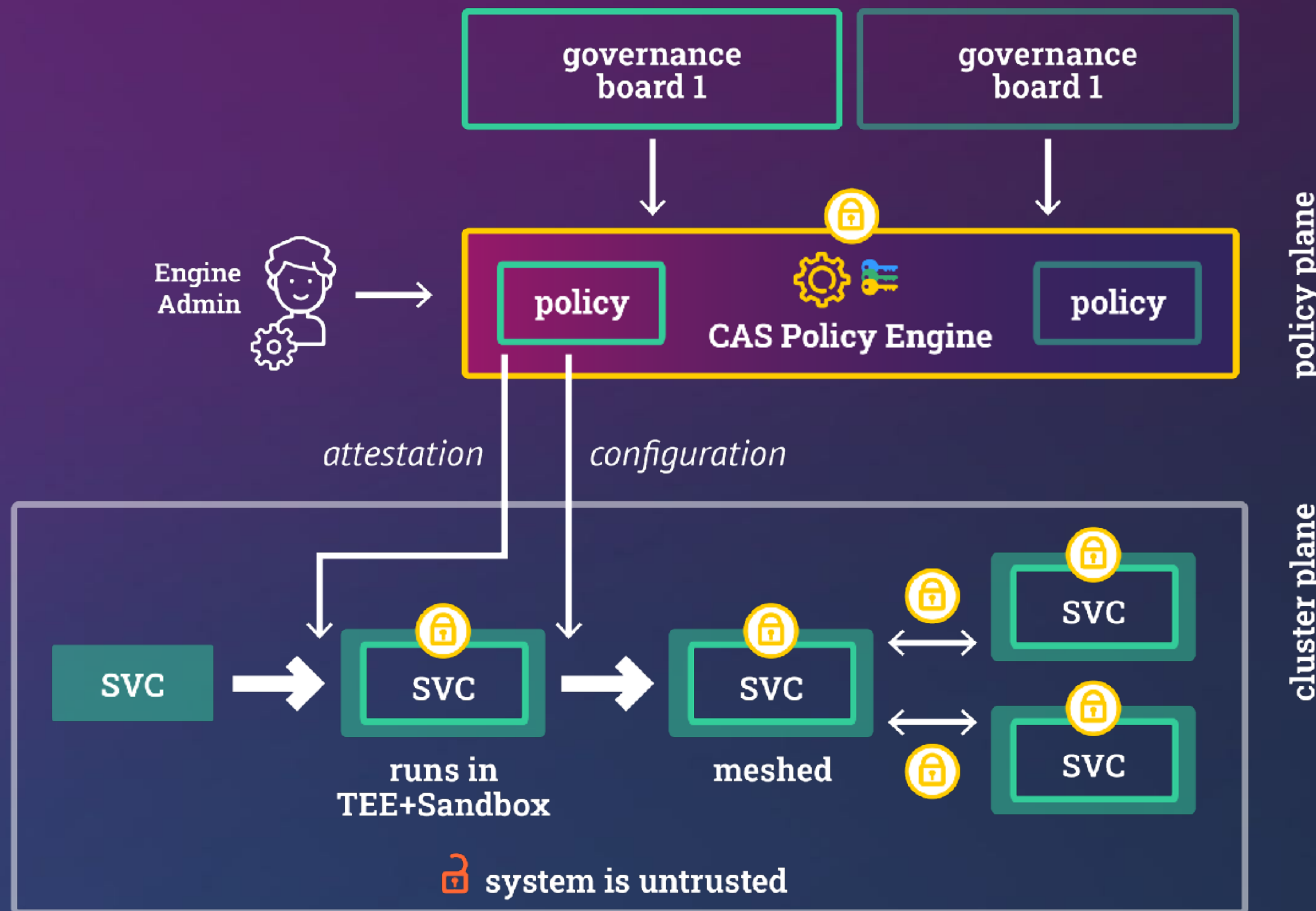




SCONE TD Build Service

SCONE Architecture



Trusted Execution Environment (TEE):



a memory region that is separated from operating system/hypervisor/DMA. It ensures data is stored, processed and protected outside accesses.

Enclave is a TEE using an encrypted memory region only the code inside the enclave can access.



Sandbox:

mechanism for separating a running program - preventing it to access the outside

Trusted Computing Base (TCB):

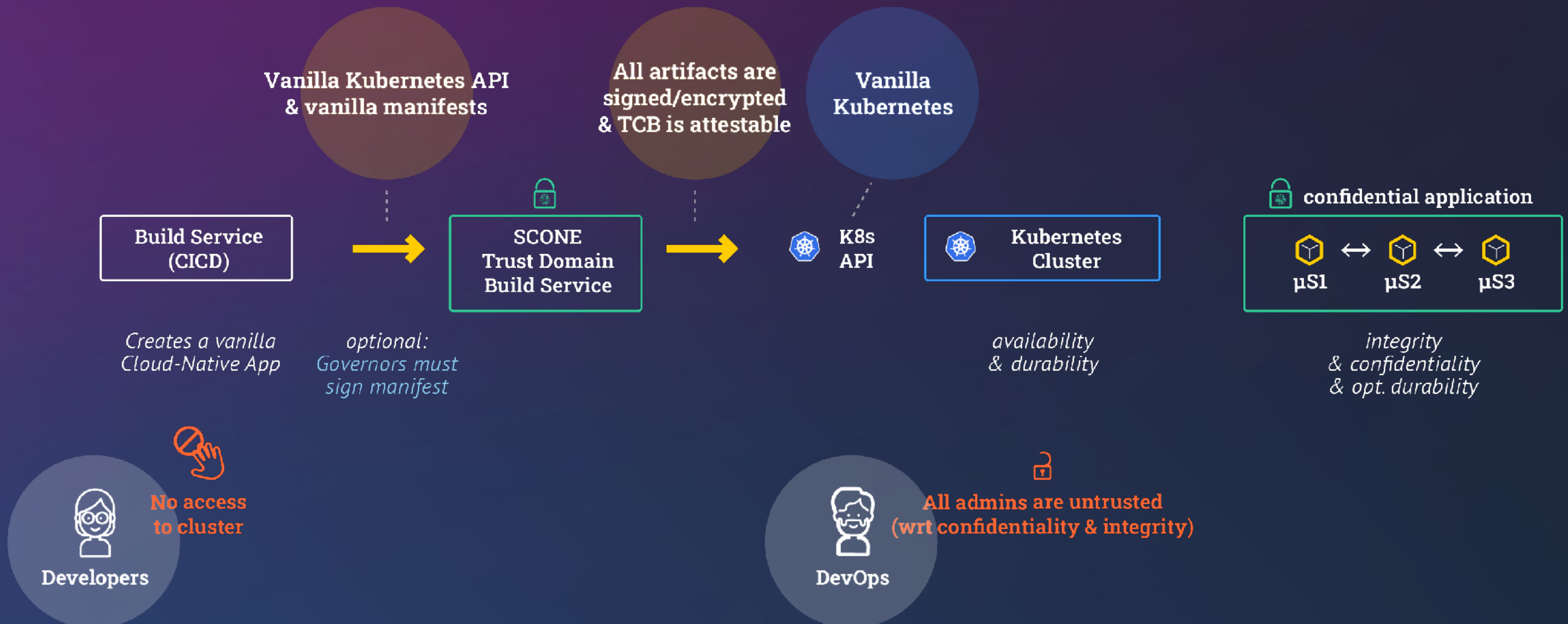


contains all components in a computing system that provides a secure environment for operations.

SCONE TD Build Service

Designtime

Runtime

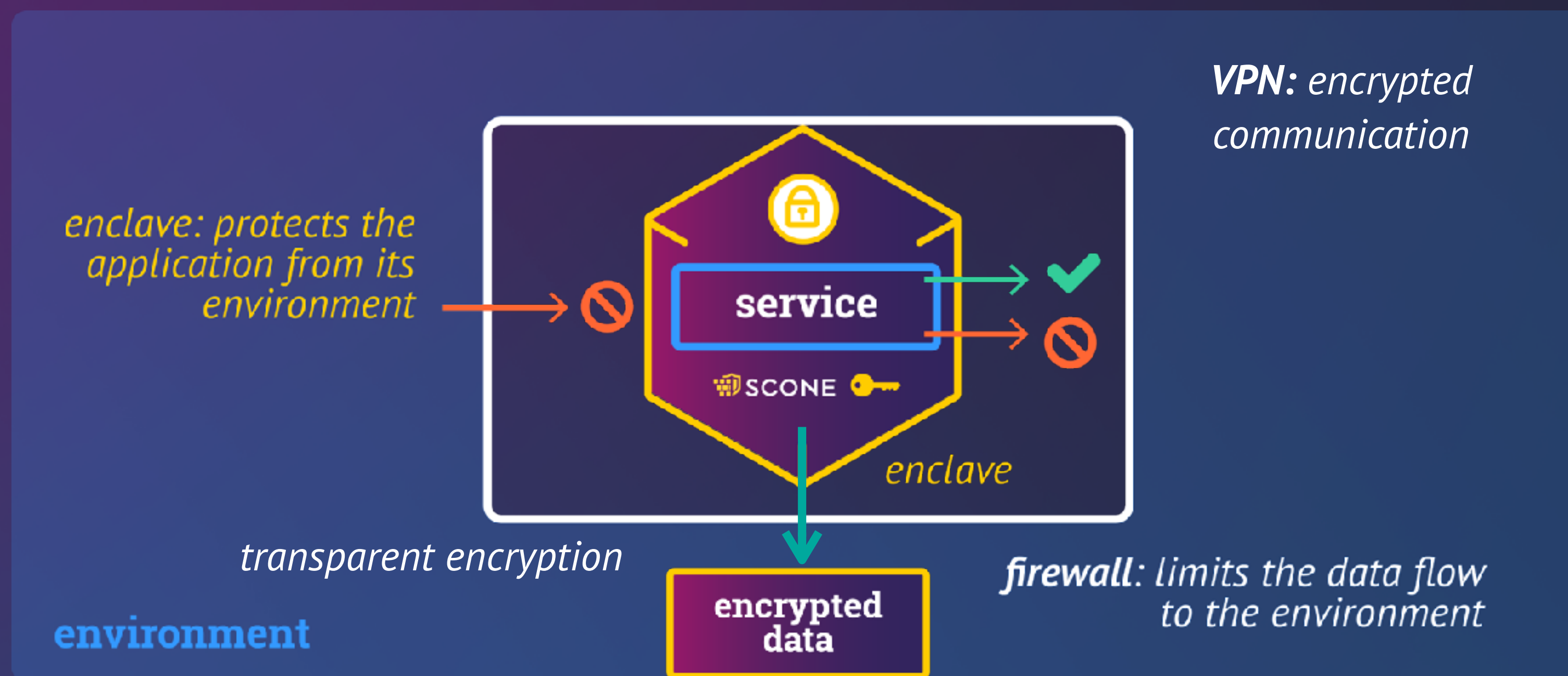


Protect the DATA, CODE & SECRETS



Protection in Use

- TEE & Sandbox -



Building An Image for Intel SGX

```
./scripts/td-build.sh
```

```
🔑 Generating confidential manifests...
```

```
🔧 Rendering confidential manifest: /home/ubuntu/GIT/java-args-env-file/scripts/../../generated/initdata.yaml
```

```
🔧 Rendering confidential manifest: /home/ubuntu/GIT/java-args-env-file/scripts/../../generated/initdata.yaml
```

```
🔧 Registering image: registry.scontain.com/workshop/java-cli-env-reader@sha256:692a7ae54943cfaeed829b2aea497196b2ef12074eeb73f973e21f131d08d2e8 as registry.scontain.com/workshop/java-cli-env-reader:692a7ae54943cfaeed829b2aea497196b2ef12074eeb73f973e21f131d08d2e8
```

```
[2025-07-20T13:21:45Z INFO k8s_scone::register] Registering image...
```

```
[2025-07-20T13:21:45Z INFO k8s_scone::register] Storage file does not exist. Initializing storage file.  
No such file or directory (os error 2)
```

```
[2025-07-20T13:21:45Z INFO k8s_scone::tools::docker]
```

```
'sha256:aaaf77ec95aa0850d2324acd318fdd74edf5bb621939a6d0c72524bb1c5d183a'
```

```
...
```

```
[2025-07-20T13:25:18Z INFO k8s_scone::apply] Session stored on ./generated/manifest.session.yaml
```

```
[2025-07-20T13:25:18Z INFO k8s_scone::apply] Manifest stored on ./generated/manifest.sanitized.yaml
```

```
[2025-07-20T13:25:18Z INFO k8s_scone::apply] The manifest processing is finished
```

```
[2025-07-20T13:25:18Z INFO k8s_scone::apply] To deploy the application apply the trust policy manifest ./generated/manifest.sanitized.yaml
```

```
[2025-07-20T13:25:18Z INFO k8s_scone::apply] To debug your corresponding policy session apply the manifest.manifest.session.yaml
```

```
[2025-07-20T13:25:18Z INFO k8s_scone::from] Unknown API version: apps/v1
```

```
✅ Confidential manifest: generated/manifest.sanitized.yaml
```

```
🎉 Confidentialization completed successfully.
```

- `./script/td-build.sh`

Running the Confidential App

```
k apply -f generated/manifest.sanitized.yaml
```

```
deployment.apps/java-cli-env-reader created
```

```
encryptedpolicy.cas.scone.cloud/encryptedpolicy-eyjv7h7ueo-session-k8s-scone created
```

```
encryptedpolicy.cas.scone.cloud/encryptedpolicy-eyjv7h7ueo-session-k8s-scone-deployment-java-cli-env-reader created
```

```
encryptedpolicy.cas.scone.cloud/encryptedpolicy-eyjv7h7ueo-session-k8s-scone-java-cli-env-secret created
```

```
encryptedpolicy.cas.scone.cloud/encryptedpolicy-eyjv7h7ueo-session-k8s-scone-java-cli-env-config created
```

```
encryptedpolicy.cas.scone.cloud/encryptedpolicy-eyjv7h7ueo-session-k8s-scone-deployment-java-cli-env-reader-config-fragments-versions created
```

```
encryptedpolicy.cas.scone.cloud/encryptedpolicy-eyjv7h7ueo-session-k8s-scone-deployment-java-cli-env-reader-java-cli-env-reader-cas-default-versions created
```

```
encryptedpolicy.cas.scone.cloud/encryptedpolicy-eyjv7h7ueo-session-k8s-scone-deployment-java-cli-env-reader-config-fragments-opt-scone-bin-rust-cli created
```

```
encryptedpolicy.cas.scone.cloud/encryptedpolicy-eyjv7h7ueo-session-k8s-scone-deployment-java-cli-env-reader-java-cli-env-reader-cas-default-opt-scone-bin-rust-cli created
```

```
encryptedpolicy.cas.scone.cloud/encryptedpolicy-eyjv7h7ueo-session-k8s-scone-deployment-java-cli-env-reader-config-fragments-versions-opt-scone-bin-rust-cli created
```

```
encryptedpolicy.cas.scone.cloud/encryptedpolicy-eyjv7h7ueo-session-k8s-scone-deployment-java-cli-env-reader-java-cli-env-reader-cas-default-versions-opt-scone-bin-rust-cli created
```

```
encryptedpolicy.cas.scone.cloud/encryptedpolicy-eyjv7h7ueo-session-k8s-scone-deployment-java-cli-env-reader-config-fragments-opt-java-openjdk-bin-java created
```

```
encryptedpolicy.cas.scone.cloud/encryptedpolicy-eyjv7h7ueo-session-k8s-scone-deployment-java-cli-env-reader-java-cli-env-reader-cas-default-opt-java-openjdk-bin-java created
```

```
encryptedpolicy.cas.scone.cloud/encryptedpolicy-eyjv7h7ueo-session-k8s-scone-deployment-java-cli-env-
```

▶ Demo: Inspecting Secrets

We can read the secrets and configurations by

- executing a shell inside the container

```
ubuntu@apple-1:java-args-env-file$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
java-cli-env-reader-554d78485c-vf2z5 1/1     Running   0           59m
...
ubuntu@apple-1:java-args-env-file$ k exec -it java-cli-env-reader-554d78485c-vf2z5 -- bash
root@java-cli-env-reader-554d78485c-vf2z5:/app# ls /config/
...
```

Inspecting Secrets & ConfigMaps

- Demo
 - inspecting ConfigMaps - none related to the application
 - inspecting Secrets - none related to the application